

SURFACE DISPLAY SELECTOR

Site Survey Protocol, Data Entry, and Photometric Decision Engine User Guide

Document Purpose

This document details the standardized field procedure for evaluating physical projection surfaces, measuring ambient lighting conditions, and executing the Surface Display Selector decision engine. It outlines mathematical formulas used to derive lumens, throws, and resolutions to ensure high-fidelity UI legibility.

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1. Instruments & Survey Protocol

Executing an accurate physical site survey is the absolute prerequisite for successful projection design. Standard room lighting and projection surface properties dictate whether projection will succeed or fail.

INSTRUMENT KIT

Every surveyor must carry and utilize the standard survey kit:

- Lux Meter: Any calibrated digital lux meter ($\pm 5\%$ accuracy). *Smartphone lux meter apps are for triage only and are prone to $\pm 30\%$ sensor errors.*
- Laser Distance Meter: Used for measuring throw distance, image height, image width, and viewing sightlines.
- Flashlight: For the surface specularity (gloss) test.
- White A4 Paper Sheet: Serving as a reference for reflectance comparison.
- Camera: Documenting context, mounting angles, and surrounding light sources.

STEP-BY-STEP PROCEDURE

1. S1: Define the Image and Viewer. Mark the target display boundaries with low-tack tape. Measure the width (W) and height (H) in meters. Measure the closest viewing distance (D) — for interactive touch systems, this is typically arm's length (0.5m - 0.7m). Define the content class carefully (Video vs. Dense UI).
2. S2: Measure Ambient Illuminance. Place the lux meter dome flat against the wall, facing out toward the room (measuring light arriving at the surface). Take 5-point readings (4 corners + center). Capture readings under worst-case scenario (e.g., mid-day sun, blinds open, overhead lights full).
3. S3: Characterise the Surface. Hold the A4 white paper (approx. 0.85 reflectance) next to the wall. Estimate surface reflectance: white (0.85), light gray/beige (0.60), mid-tone (0.35), or dark/brick (0.15). Conduct the *specularity test* by shining the flashlight at a 45° angle to identify glare hotspots.

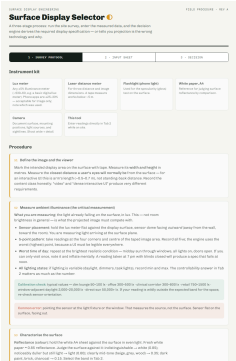
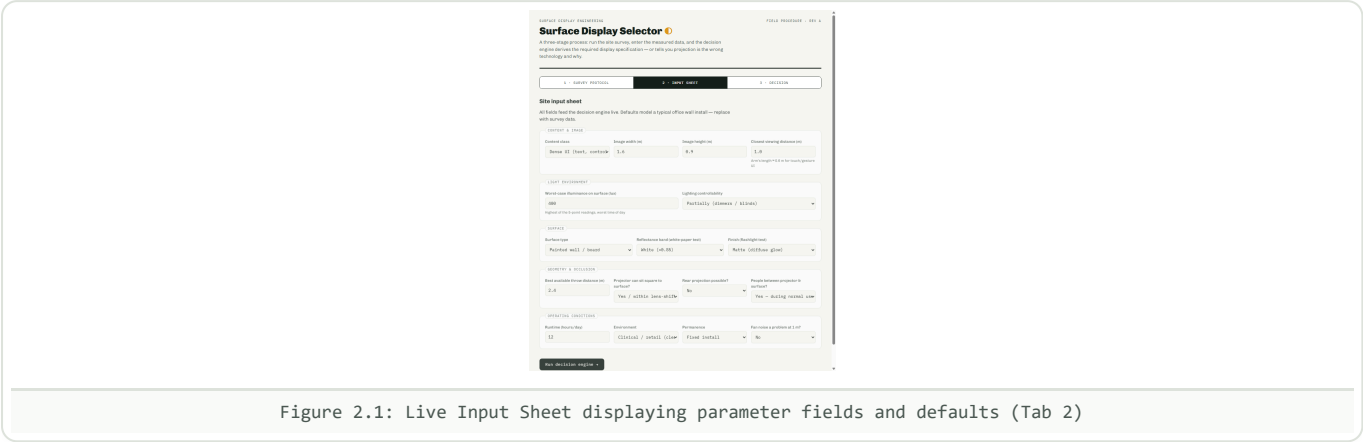


Figure 1.1: Surface Display Selector - Step-by-Step Field Survey Protocol (Tab 1)

2. Input Sheet Parameters

The measured site parameters are entered directly into Tab 2: Input Sheet. These inputs drive the decision engine rules and calculate requirements in real-time.



PARAMETERS & GUIDELINES

PARAMETER	DEFAULT VALUE	ENGINEERING GUIDANCE
Content Class	Dense UI	Select 'Video' for moving imagery, 'Signage' for bold graphics, or 'Dense/Interactive UI' for text-heavy layouts and interactive targets.
Image Dimensions	1.6m × 0.9m	Physical dimensions of the projected area. W and H determine total area A.
Worst-Case Illuminance	400 lux	The maximum lux value recorded in the 5-point survey. Office environments typically range from 300 to 500 lx.
Surface Reflectance	White (0.85)	p value. Lower reflectance absorbs more projected light, requiring a brighter projector.
Occlusion Risk	Yes	Set to 'Yes' if users will stand between the projector and wall. This triggers the mandatory UST (Ultra-Short-Throw) requirement.

Calibration Check: If you record an ambient illuminance of 0 lux or > 20,000 lux (unless in direct outdoors sun), check sensor calibration. Always multiply foot-candle readings by 10.76 to convert to lux.

3. Decision Engine & Photometrics

The decision engine processes inputs dynamically, evaluating two primary mathematical constraints to derive the necessary luminous flux (ANSI lumens) and display properties.

Lumen Calculation Formulas:

- 1. Contrast Path: $\Phi_{\text{contrast}} = (C_{\text{target}} - 1) \cdot E \cdot A$
- 2. Absolute Path: $\Phi_{\text{absolute}} = L_{\text{min}} \cdot \pi \cdot A / \rho$
- 3. Binding Lumen: $\Phi_{\text{on-screen}} = \max(\Phi_{\text{contrast}}, \Phi_{\text{absolute}})$
- 4. Spec Requirement: $\Phi_{\text{spec}} = \Phi_{\text{on-screen}} \cdot 1.5$ [Headroom: 1.25x real-vs-spec, 1.2x lamp aging]

PARAMETERS FOR CONTENT CLASSES:

- Video: $C_{\text{target}} = 8, L_{\text{min}} = 100$ nits
- Signage: $C_{\text{target}} = 6, L_{\text{min}} = 200$ nits
- Dense UI / Interactive: $C_{\text{target}} = 15, L_{\text{min}} = 150$ nits

RESOLUTION ACUITY LIMITS:

Acuity limit $p = D \times 0.00029$ (representing 1 arcminute of resolution at viewing distance D). For UI content, the system enforces a pixel size $\leq p$. The minimum horizontal resolution is calculated as $\text{ceil}(W / p)$.

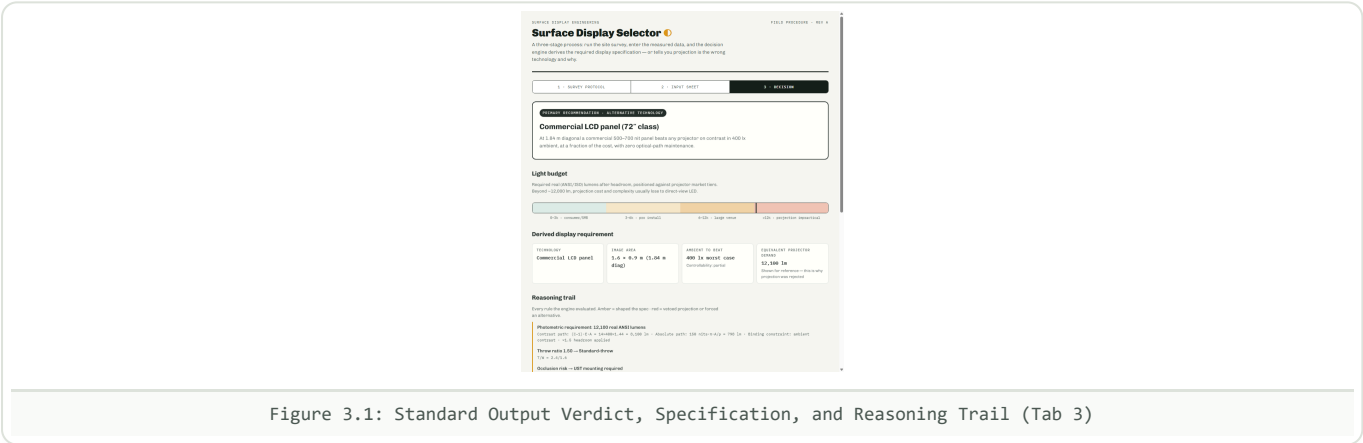


Figure 3.1: Standard Output Verdict, Specification, and Reasoning Trail (Tab 3)

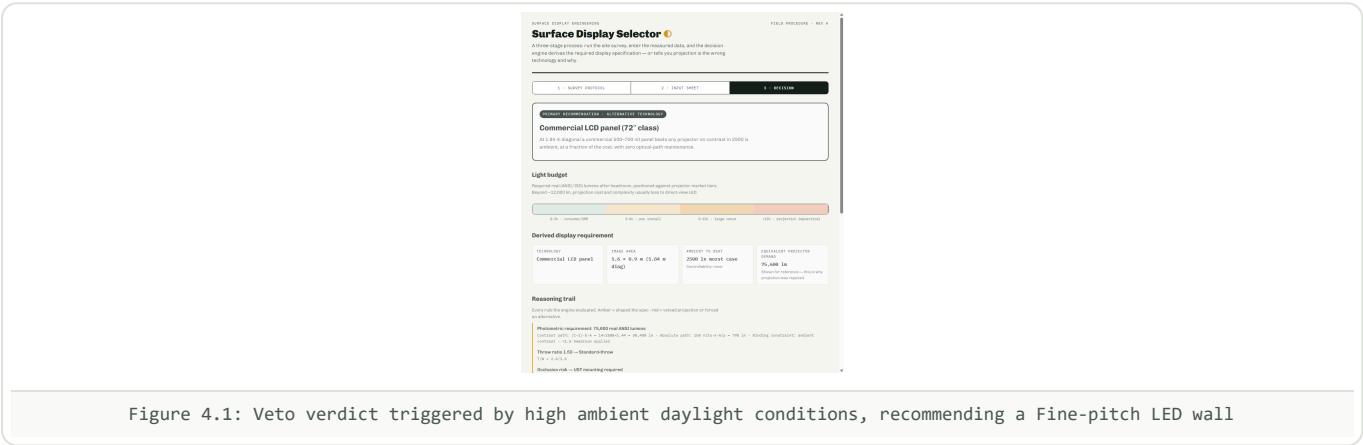
Throw Ratio (TR): Classifies optics based on T/W. Standard ranges: UST (<0.4), Short-throw (0.4–1.0), Standard (1.0–2.1), and Long-throw (>2.1).

4. Veto Conditions & Alternatives

Certain physical constraints make projection non-viable. The engine triggers vetoes under these conditions, highlighting them in red on the reasoning trail and advising alternative displays.

VETO RULES AND TRIGGERS

- Daylight + UI: Uncontrolled daylight with interactive UI forces alternative tech, as legibility cannot be guaranteed under variable sunlight.
- Lumen Ceiling: When calculated requirement exceeds 12,000 ANSI lumens. Large venue projection becomes economically and logistically impractical compared to direct-view emissive displays.
- Glass without Rear Access: Front projection fails on transparent glass because light passes straight through rather than diffusing.



ALTERNATIVE TECHNOLOGY FITS

- Commercial LCD Panel: Best fit when image diagonal is ≤ 2.4 meters. Offers much higher contrast, zero optical path maintenance, and better UI clarity in office lighting.
- Fine-Pitch LED Wall: Best fit when calculated lumens or image size exceed projection capabilities. Modular tiles are daylight-proof but demand structural support.
- Transparent LED Film / Rear Projection Film: Standard alternatives when glass surfaces are used. Rear film requires a rear cavity throw path.